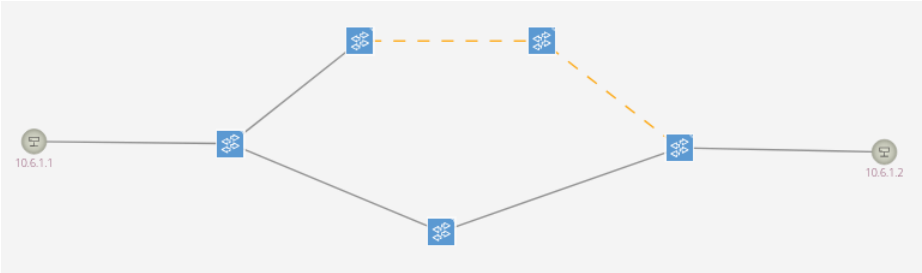
Monitor s1 and s4 interface

```
mininet> h2 iperf -s -u &
mininet> h1 iperf -c 10.6.1.2 -u -b 1M -t 5
------------------------------------------------------------
Client connecting to 10.6.1.2, UDP port 5001
Sending 1470 byte datagrams, IPG target: 11215.21 us (kalman adjust)
UDP buffer size:  208 KByte (default)
------------------------------------------------------------
[  1] local 10.6.1.1 port 60760 connected with 10.6.1.2 port 5001
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-5.0136 sec   645 KBytes  1.05 Mbits/sec
[  1] Sent 450 datagrams
[  1] Server Report:
[ ID] Interval       Transfer     Bandwidth        Jitter   Lost/Total Datagrams
[  1] 0.0000-4.7758 sec   613 KBytes  1.05 Mbits/sec   0.020 ms 22/449 (4.9%)
mininet> sh ovs-ofctl dump-ports -O OpenFlow14 s1
OFPST_PORT reply (OF1.4) (xid=0x2): 4 ports
  port LOCAL: rx pkts=0, bytes=0, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=0, bytes=0, drop=0, errs=0, coll=0
           duration=82.939s
  port  "s1-eth1": rx pkts=910, bytes=695122, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=82, bytes=10795, drop=0, errs=0, coll=0
           duration=82.954s
  port  "s1-eth2": rx pkts=79, bytes=10493, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=977, bytes=704647, drop=0, errs=0, coll=0
           duration=82.958s
  port  "s1-eth3": rx pkts=82, bytes=10941, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=79, bytes=10493, drop=0, errs=0, coll=0
           duration=82.954s
mininet> sh ovs-ofctl dump-ports -O OpenFlow14 s4
OFPST_PORT reply (OF1.4) (xid=0x2): 3 ports
  port LOCAL: rx pkts=0, bytes=0, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=0, bytes=0, drop=0, errs=0, coll=0
           duration=86.191s
  port  "s4-eth1": rx pkts=81, bytes=10771, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=84, bytes=11219, drop=0, errs=0, coll=0
           duration=86.202s
  port  "s4-eth2": rx pkts=83, bytes=11031, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=83, bytes=11049, drop=0, errs=0, coll=0
           duration=86.201s
```

從圖中可以看到 h1 iperf 會因為 group 而從 s1 port2 傳送。

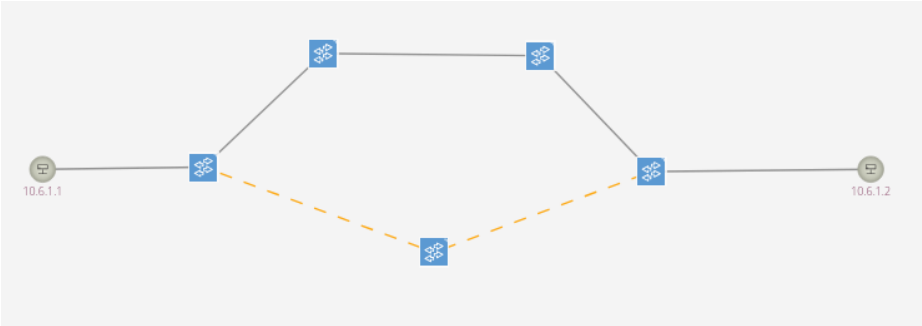
- Check if traffic from h1 to h2 through s2 and s3

會透過 intent 經過 s2, s3 到達 s5



- Check the path from h2 to h1 through which switch

會透過 intent 經過 s4 到達 s1



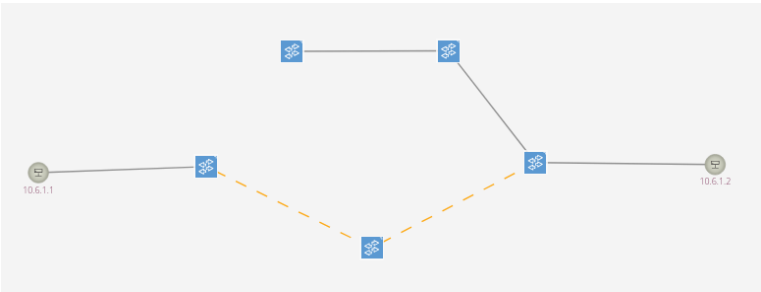
Monitor s1 and s4 interface

```
mininet> link s1 s2 down
mininet> h1 iperf -c 10.6.1.2 -u -b 1M -t 5
------------------------------------------------------------
Client connecting to 10.6.1.2, UDP port 5001
Sending 1470 byte datagrams, IPG target: 11215.21 us (kalman adjust)
UDP buffer size:  208 KByte (default)
------------------------------------------------------------
[  1] local 10.6.1.1 port 35123 connected with 10.6.1.2 port 5001
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-5.0142 sec   645 KBytes  1.05 Mbits/sec
[  1] Sent 450 datagrams
[  1] Server Report:
[ ID] Interval       Transfer     Bandwidth        Jitter   Lost/Total Datagrams
[  1] 0.0000-4.7761 sec   370 KBytes   635 Kbits/sec   0.018 ms 191/449 (43%)
mininet> sh ovs-ofctl dump-ports -O OpenFlow14 s1
OFPST_PORT reply (OF1.4) (xid=0x2): 4 ports
  port LOCAL: rx pkts=0, bytes=0, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=0, bytes=0, drop=0, errs=0, coll=0
           duration=50.354s
  port  "s1-eth1": rx pkts=912, bytes=696688, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=60, bytes=7838, drop=0, errs=0, coll=0
           duration=50.370s
  port  "s1-eth2": rx pkts=33, bytes=4301, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=33, bytes=4301, drop=0, errs=0, coll=0
           duration=50.372s
  port  "s1-eth3": rx pkts=58, bytes=7706, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=955, bytes=702958, drop=0, errs=0, coll=0
           duration=50.369s
mininet> sh ovs-ofctl dump-ports -O OpenFlow14 s4
OFPST_PORT reply (OF1.4) (xid=0x2): 3 ports
  port LOCAL: rx pkts=0, bytes=0, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=0, bytes=0, drop=0, errs=0, coll=0
           duration=53.212s
  port  "s4-eth1": rx pkts=957, bytes=703236, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=60, bytes=7984, drop=0, errs=0, coll=0
           duration=53.226s
  port  "s4-eth2": rx pkts=58, bytes=7706, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=787, bytes=455826, drop=0, errs=0, coll=0
           duration=53.226s
```

從圖中可以看到在 s1 s2 link down 之後，h1 iperf 會因為 group 而從 s1 port3 傳送。

- Check if both traffic go through s4

都會透過 intent 從 s4 傳送



- Check if the iperf traffic rate is limited

可以看到隨著 bandwidth 提高的時候 lost data 也會隨之提升，從 512K 的時候是 0%到 1M 的時候變成 45%，也就是受到 meter 的限制，而在超過 burstsize1M 到 5M 的時候又會大幅提升到 92%。

```
mininet> h1 iperf -c 10.6.1.2 -u -b 512K -t 5
------------------------------------------------------------
Client connecting to 10.6.1.2, UDP port 5001
Sending 1470 byte datagrams, IPG target: 22430.42 us (kalman adjust)
UDP buffer size:  208 KByte (default)
------------------------------------------------------------
[  1] local 10.6.1.1 port 58315 connected with 10.6.1.2 port 5001
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-5.0303 sec   324 KBytes   528 Kbits/sec
[  1] Sent 227 datagrams
[  1] Server Report:
[ ID] Interval       Transfer     Bandwidth        Jitter   Lost/Total Datagrams
[  1] 0.0000-5.0247 sec   324 KBytes   529 Kbits/sec   0.022 ms 0/226 (0%)
mininet> h1 iperf -c 10.6.1.2 -u -b 1M -t 5
------------------------------------------------------------
Client connecting to 10.6.1.2, UDP port 5001
Sending 1470 byte datagrams, IPG target: 11215.21 us (kalman adjust)
UDP buffer size:  208 KByte (default)
------------------------------------------------------------
[  1] local 10.6.1.1 port 46186 connected with 10.6.1.2 port 5001
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-5.0133 sec   645 KBytes  1.05 Mbits/sec
[  1] Sent 450 datagrams
[  1] Server Report:
[ ID] Interval       Transfer     Bandwidth        Jitter   Lost/Total Datagrams
[  1] 0.0000-5.0502 sec   392 KBytes   636 Kbits/sec   2.330 ms 177/450 (39%)
mininet> h1 iperf -c 10.6.1.2 -u -b 5M -t 5
------------------------------------------------------------
Client connecting to 10.6.1.2, UDP port 5001
Sending 1470 byte datagrams, IPG target: 2243.04 us (kalman adjust)
UDP buffer size:  208 KByte (default)
------------------------------------------------------------
[  1] local 10.6.1.1 port 56205 connected with 10.6.1.2 port 5001
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-5.0024 sec  3.13 MBytes  5.25 Mbits/sec
[  1] Sent 2233 datagrams
[  1] Server Report:
[ ID] Interval       Transfer     Bandwidth        Jitter   Lost/Total Datagrams
[  1] 0.0000-5.0242 sec   243 KBytes   396 Kbits/sec   1.383 ms 2064/2233 (92%)
```